

1. Executive Summary & Core Objective

Aria is engineered as an enterprise-grade finance platform designed to automate full, end-to-end finance processes. It explicitly rejects the paradigm of isolated point solutions or fragile, surface-level workflow automation. The core objective of Aria is to encode domain judgment, financial controls, evidentiary audit trails, and deterministic exception handling into repeatable, cross-functional operating systems.

Every functional implementation inside this architecture strictly isolates the commercial product modules from the underlying reusable capability layers (the data engines, policy models, and transactional rails). One-off client builds are prohibited; every engineering cycle must deliver platform foundations that compound value across the entire finance suite.

2. Platform Capability Framework (The Engine)

The runtime environment instantiates seven core reusable capability layers. These serve as the shared engine underneath all commercial modules:

- **Source Data Connectors & Evidence Graph:** Unified data ingestion pipelines that preserve complete data lineage from underlying ERP systems (specifically SAP core tables) and construct verifiable typographic links between records.
- **Policy-as-Code / Rule Engine:** A deterministic execution environment where corporate finance policies, treasury guidelines, and compliance constraints are converted into executable logic patterns.
- **Domain Judgment Models:** Machine learning architectures trained to parse nuanced fact patterns against codified technical accounting rules, documenting a defensible rationale before proposing data adjustments.
- **Exception Taxonomy & Routing:** A standardized structure for diagnosing structural or operational system anomalies and routing them directly to active human-in-the-loop (HITL) triage queues.
- **Reviewer Gates & Approvals:** Cryptographically validated approval loops logging explicit enterprise authorizations prior to executing material changes in systems of record.
- **Workpaper / Memo / Report Generation:** Deterministic document engines pulling system data lineages to compile audit-ready memos, schedules, and compliance disclosures.
- **Scenario & Assumption Engine:** A localized multi-variable simulation workspace allowing corporate teams to calculate balance sheet and liquidity impact under varying operational parameters.

3. Commercial Product Catalog Mapping

Bespoke client configurations are mapped directly into eight distinct, sellable product lines. Each line utilizes the core capability layers described above:

Product Suite	Functional Scope	Target Live Client Wedges
1. Revenue Lifecycle Intelligence	Offer-to-bill pipelines, billing logic verification, contract revenue mapping, and leakage protection.	TIM, Amplify

Product Suite	Functional Scope	Target Live Client Wedges
2. Working Capital & Cash Conversion	Receivables optimization, payables smoothing, dynamic collections, and cash position forecasting.	TIM
3. Close & Controllershship Command Center	Close task automation, reconciliations, journal entries, flux analysis, and review queue monitoring.	ASML, Meta
4. External Reporting + Stat Reporting	Financial statement table compilation, footnote tie-outs, US GAAP to IFRS local bridge automation.	ASML, Meta
5. ICFR / SOX 404 Process Intelligence	Risk identification, design effectiveness verification, control execution tracking, and automated testing.	Various, Meta, ASML
6. Technical Accounting & Judgment	Fact pattern analysis, memo drafting, contract interpretation, and disclosure downstream linkages.	ASML, Amplify
7. FP&A Decision Intelligence	Ad-hoc corporate data requests, budgeting replacement workflows, and executive strategic modeling.	All Accounts
8. Treasury & Liquidity Intelligence	Daily positioning, payment control validation, currency risk forecasting, and covenant monitoring.	TIM

4. Blueprint: Working Capital Forecasting & Treasury Cash Conversion (The TIM Wedge)

This implementation specification instantiates a closed-loop system for Product Suite 2 and 8, leveraging an active integration with an SAP system to drive cash optimization outcomes.

A. Core Objective Function

The system executes an optimization routine to maximize free cash flow velocity and predictability. It continuously evaluates cash balances against outgoing payment tracks, identifying impending liquidity tightening and running micro-adjustments on settlement rules without damaging core supplier relationships.

B. SAP Information Retrieval Protocol

The data agent maps low-latency relational extractions across the following core SAP database structures:

- **BSIK** (Accounts Payable: Open Items): Isolates vendor liabilities, dates, currencies, and documents.
- **BSID** (Accounts Receivable: Open Items): Parses open client obligations to track cash receipt vectors.
- **T052** (Terms of Payment Master): Evaluates early payment baseline terms and explicit discount options.
- **ACDOCA** (Universal Journal Ledger): Processes historical transactional settlement lags to weigh actual customer behavior.

```
-- Automated SAP Context Extraction Context
SELECT bsik.bukrs AS company_code, bsik.lifnr AS vendor_id, bsik.belnr AS doc_number,
       bsik.wrbtr AS amount_foreign, bsik.waers AS currency, bsik.bline AS baseline_date,
       bsik.zterm AS pay_terms, t052.ztag1 AS discount_days_1, bsik.zlspr AS payment_block
FROM bsik
INNER JOIN t052 ON bsik.zterm = t052.zterm
WHERE bsik.zlspr = ' ';
```

C. Operational Decision & Recommendation Engine

Upon detecting an unfavorable cash trend line in the upcoming 30-day forecast, Aria engages its Policy-as-Code engine to evaluate optimization routes against strict corporate rules (e.g., minimum cash buffers). It generates two transactional recommendation arrays:

- **Inbound Acceleration:** Modifies customer parameters to activate minor discounts and pull in key receivables.
- **Outbound Smoothing:** Modifies payment baseline dates on non-discount vendor records to safely extend DPO.

D. Closed-Loop SAP Write-Back Specification

Following an explicit confirmation through the secure Reviewer Gate, the Antigravity system executes transactional modifications directly back into the SAP runtime environment via standard Business Application Programming Interfaces (BAPIs).

BAPI Interventions

1. Accounts Payable Document Adjustment (Extending DPO Limits)

Modifies the payment baseline parameters or sets specific document blocks to adjust cash run processing.

```
{
  "BAPI_ACC_DOCUMENT_CHANGE": {
    "DOCUMENTHEADER": { "OBJ_TYPE": "BKPFF", "COMPANY_CODE": "1000", "DOC_NO":
"19000004121", "FISCAL_YEAR": "2026" },
    "CRITERIA": { "ITEM_NO": "001", "FIELD_NAME": "ZFB DT", "FIELD_VALUE": "20260615" }
  }
}
```

2. Accounts Receivable Master Modification (Inbound Inflow Optimization)

Temporarily shifts customer profile terms to trigger rapid settlement cycles for active outstanding accounts.

```
{
  "BAPI_CUSTOMER_EXTENS_CHG": {
    "CUSTOMERADDRESS": { "CUSTOMER": "0000401290" },
    "COMPANYDATA": { "COMPANY_CODE": "1000", "PAY_TERMS": "Z010" }
  }
}
```

E. Verification, Lineage & Roadmap Milestones

Post-execution, Aria re-queries the modified tables to guarantee the updates match. It bundles the source records, rule clearances, and logs into a single audit packet. Development follows the compressed 90-day cycle: Weeks 0-2 for scoping data contracts, Weeks 3-6 for active SAP CAL demos, Weeks 7-10 for pilot deployment, and Weeks 11-16 for shipping a fully commercialized package.